
Data 250 Practice Exam 1

1. For $n=0$ to $n=3$, calculate $a_n = \lfloor 1.6n \bmod (5 \lceil \pi \rceil, 6) \rfloor$

Notation: $\lfloor x \rfloor$ is the floor of x and $\lceil x \rceil$ is the ceiling of x

2. Find a recurrence for the number of strings containing only A or B with no substrings AAB.

3. Find a closed form for

$$Out[i,j] = \sum_{i=1}^n \sum_{j=1}^n j \cdot 3^i$$

4. Give a truth table for

if $((p \text{ and } q) \text{ xor } q)$:

p

else:

q

5. Give a table of $x^2 \bmod 4$ where x ranges from 0 to 3.